

AVINASH SINGH

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Professional Experience

Software Engineer, Data & AI Platforms | Visa Inc.

Apr 2025 – Present

- Led end-to-end development of Unified Access Manager, a governance platform standardizing access control across AWS, Databricks & on-premises Hadoop systems, securing enterprise-wide data sharing across **3000+ databases**.
- Architected the core policy evaluation engine & authorization workflow, enabling fine-grained permissions at table, row, & column levels across **600K+ tables**, ensuring consistency, compliance, & secure enforcement in distributed data environments.
- Optimized policy execution pipeline to process **2M+ access rules** with **p99 latency under 50ms**, improving enforcement reliability for Visa's global data operations.
- Integrated with three governance systems (**Unity Catalog, Lake Formation, Ranger**) using RESTful APIs & event-driven architecture, enabling seamless cross-platform policy synchronization & centralized audit logging.
- Re-engineered the Active Directory sync & lookup jobs for **50K+ AD group members**, introducing connection pooling, batched LDAP queries & retry-safe execution, cutting end-to-end job runtime from **30–40 minutes to under 30 seconds** while improving reliability & eliminating partial-sync failures.
- Drove code quality & release health, by resolving **10+ SonarQube & Sonatype** findings, maintained **85% test coverage** with **less than 0.5% duplication**, and remediated **20+ legacy Checkmarx vulnerabilities** including **SQL injection, parameter tampering, frameable login page & privacy violations**; managed security gates, containerization & deployment.

Product Solutions Engineer (Backend) | PhonePe

Jul 2023 – Mar 2025

- Architected & delivered a highly scalable Agent Console Platform, enabling real-time ticket routing & adaptive PII masking utilizing event-driven processing, **supporting 5000+ agents & 500K+ daily interactions with 99.95% availability**.
- Designed & implemented a rule-driven escalation & SLA orchestration engine with configurable routing logic, automated prioritization, & failure-safe retries, **cutting average resolution time by 35%** & raising first-contact resolution rates across support tiers.
- Architected Agent Assistance system with MCP servers integrating GenAI & in-house models, providing real-time response suggestions & customer context retrieval that **reduced average handle time by 25%** across 2M+ monthly conversations.
- Engineered customer chatbot backend with horizontal auto-scaling supporting PhonePe's customer service operations with **20K+ concurrent sessions**, integrated with **20+ backend services** via async APIs with **p95 latency under 200ms**.
- Developed distributed caching strategy with Redis clustering & TTL-based invalidation, reducing **database load by 60%** & improving response times for frequently accessed customer data across microservices.

Graduate Trainee - Intern | PhonePe

Jan 2023 – Jun 2023

- Designed & implemented an RBAC Console with hierarchical role inheritance & in-memory caching, integrating seamlessly **across 20+ microservices**.
- Enabled secure authorization for **5,000+ users** with **sub-10ms** permission checks & 99.99% availability, improving access governance & system reliability.

Skills

Backend & Frameworks: Java, Python, Spring Boot, Spring WebFlux

Architecture & Design: Microservices, Event-Driven Systems, Distributed Caching, High Availability, Fault Tolerance

Messaging & Streaming: Apache Kafka, RabbitMQ, AWS SNS/SQS

Databases & Caching: MySQL, PostgreSQL, Redis, Memcached, Elasticsearch

Cloud & Data Platforms: AWS (EC2, ECS/EKS, S3, Lambda, Amplify, EventBridge, CloudWatch, CloudTrail, SNS, SQS, ALB, ASG, Glue, Athena, Lake Formation), Databricks (Unity Catalog, Delta Lake, Delta Live Tables, Lakehouse Federation)

DevOps & Security: Linux, Git, Docker, Kubernetes, Jenkins, GitHub Actions, SAST/SCA (SonarQube, Checkmarx, Sonatype)

Certifications

Redis for Java Developers by Redis University 

Jan 2026

Databricks Certified Data Engineer Associate 

Oct 2025

Education

National Institute of Technology (NIT), Patna

Jul 2019 – Jun 2023

Bachelor of Technology in Civil Engineering, **CGPA: 8.20/10.0**